

Scalability & Future Plan

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Team ID	SWTID-2026-3658
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the EduGenie platform can be scaled to support a larger number of users, handle increased AI requests, and incorporate advanced learning features. It also presents the roadmap for enhancing the system into a more intelligent, cloud-ready educational platform.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address
1	Limited support for concurrent users	Response time may increase during heavy usage	High
2	SQLite database is suitable for small-scale deployment only	Limited scalability for large volumes of user data	High
3	No support for document-based learning or semantic search	Users cannot upload study materials for AI-assisted learning	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a limited number of users on a Local FastAPI	Deploy on cloud infrastructure (AWS, Google Cloud, Azure) with load balancing and auto-scaling
2	Data Storage	SQLite database for local data storage	Implement scalable databases such as MongoDB or PostgreSQL with cloud storage support
3	Performance	AI requests are processed individually in real time	Use caching mechanisms, asynchronous processing, and optimized AI inference pipelines
4	Security	JWT authentication and bcrypt password hashing	Implement OAuth, encryption, role-based access control, and secure API gateways

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Cloud deployment using Docker, Google Cloud, or AWS with PostgreSQL/MongoDB	2 Months	Improves scalability, reliability, and supports more concurrent users
Phase 3	Retrieval-Augmented Generation (RAG), PDF upload, ChromaDB/FAISS integration, and multilingual learning support	4 Months	Enables personalized learning from uploaded documents and improves AI response quality
Phase 4	Mobile application, voice-based interaction, analytics dashboard, and offline AI support using Ollama	6 Months	Enhances accessibility, user engagement, and provides an enterprise-level learning experience